wdq5fxdkl

April 27, 2026

[82]:
```python
import pandas as pd
import numpy as np
```

[83]:
```python
dfgames = pd.read_csv(
    "games.csv",
    sep=",",
    encoding="utf-8",
    quotechar='"',
    engine="python",
    on_bad_lines="skip"
)

dfgames2 = pd.read_csv("games2.csv")
```

[84]:
```python
print("Columnas del dataset:")
print(dfgames.columns)

print("\nTipos de datos:")
print(dfgames.dtypes)
```

```
Columnas del dataset:
Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
       'Required age', 'Price', 'DiscountDLC count', 'About the game',
       'Supported languages', 'Full audio languages', 'Reviews',
       'Header image', 'Website', 'Support url', 'Support email', 'Windows',
       'Mac', 'Linux', 'Metacritic score', 'Metacritic url', 'User score',
       'Positive', 'Negative', 'Score rank', 'Achievements', 'Recommendations',
       'Notes', 'Average playtime forever', 'Average playtime two weeks',
       'Median playtime forever', 'Median playtime two weeks', 'Developers',
       'Publishers', 'Categories', 'Genres', 'Tags', 'Screenshots', 'Movies',
       'Column1'],
      dtype='str')

Tipos de datos:
AppID                           int64
Name                              str
Release date                      str
Estimated owners                  str
```

```
Peak CCU                        int64
Required age                    int64
Price                         float64
DiscountDLC count               int64
About the game                  int64
Supported languages               str
Full audio languages              str
Reviews                           str
Header image                      str
Website                           str
Support url                       str
Support email                     str
Windows                           str
Mac                              bool
Linux                            bool
Metacritic score                 bool
Metacritic url                  int64
User score                        str
Positive                        int64
Negative                        int64
Score rank                      int64
Achievements                  float64
Recommendations                 int64
Notes                           int64
Average playtime forever          str
Average playtime two weeks      int64
Median playtime forever         int64
Median playtime two weeks       int64
Developers                      int64
Publishers                        str
Categories                        str
Genres                            str
Tags                              str
Screenshots                       str
Movies                            str
Column1                       float64
dtype: object
```

[85]:
```
print("\nValores nulos por columna:")
print(dfgames.isnull().sum())
```

```
Valores nulos por columna:
AppID                             0
Name                              1
Release date                      0
Estimated owners                  0
Peak CCU                          0
```

```
Required age                      0
Price                             0
DiscountDLC count                 0
About the game                    0
Supported languages            8449
Full audio languages              0
Reviews                           0
Header image                 110541
Website                          81
Support url                   72935
Support email                 68469
Windows                       22263
Mac                               0
Linux                             0
Metacritic score                  0
Metacritic url                    0
User score                   118355
Positive                          0
Negative                          0
Score rank                        0
Achievements                 122571
Recommendations                   0
Notes                             0
Average playtime forever     100153
Average playtime two weeks        0
Median playtime forever           0
Median playtime two weeks         0
Developers                        0
Publishers                     8437
Categories                     8909
Genres                         8953
Tags                           8413
Screenshots                   39265
Movies                         6018
Column1                      122611
dtype: int64
```

[86]:
```
dfgames.sample(5)
```

[86]:
```
         AppID                      Name  Release date Estimated owners  \
81309  1862090        Buggy Game Playtest  Dec 31, 2021            0 - 0
11036   539750                    Kombine  Oct 14, 2016        0 - 20000
58716  1620640               Noctambulant  Aug 22, 2021        0 - 20000
52895  2104890                  RoadCraft  May 20, 2025  200000 - 500000
96399  2678640  Journey of Realm Dawn Dew  Aug 13, 2025        0 - 20000

       Peak CCU  Required age  Price  DiscountDLC count  About the game  \
```

```
81309        0             0   0.00                 0               0
11036        0             0   0.99                 0               0
58716        0             0   1.29                 0               0
52895     8156             0  27.99                30               5
96399        0             0   9.23                16               0

                                   Supported languages  …  \
81309                                               NaN  …
11036  Kombine is a minimalist puzzle game played on …  …
58716  An unfortunate event took the lives of Renee's…  …
52895  As the leader of a company specialized in rest…  …
96399   Persisting in kindness, some people say it is…  …

      Median playtime forever Median playtime two weeks Developers  \
81309                       0                        0          0
11036                       0                      277          0
58716                       0                        0          0
52895                     559                      665        930
96399                       0                      259          0

             Publishers           Categories  \
81309               NaN                  NaN
11036    Robert Alvarez       Robert Alvarez
58716         Justus Yew          Jarze Games
52895  Saber Interactive  Focus Entertainment
96399     CongSe Studio         Anotherindie

                                                 Genres  \
81309                                               NaN
11036  Single-player,Steam Achievements,Steam Trading…
58716                      Single-player,Family Sharing
52895  Single-player,Multi-player,Co-op,Online Co-op,…
96399  Single-player,Steam Achievements,Steam Trading…

                         Tags  \
81309                     NaN
11036    Casual,Indie,Strategy
58716  Adventure,Indie,Strategy
52895     Adventure,Simulation
96399       Indie,RPG,Strategy

                                            Screenshots  \
81309                                               NaN
11036  Puzzle,2D,Minimalist,Match 3,Singleplayer,Casu…
58716  Indie,Horror,Stealth,Survival Horror,Puzzle,At…
52895  Simulation,Building,Driving,Physics,Adventure,…
96399                                               NaN
```

```
                                                   Movies  Column1
81309                                                 NaN      NaN
11036  https://shared.akamai.steamstatic.com/store_it…      NaN
58716  https://shared.akamai.steamstatic.com/store_it…      NaN
52895  https://shared.akamai.steamstatic.com/store_it…      NaN
96399  https://shared.akamai.steamstatic.com/store_it…      NaN

[5 rows x 40 columns]
```

[87]:
```
dfgames.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 122611 entries, 0 to 122610
Data columns (total 40 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   AppID                       122611 non-null  int64
 1   Name                        122610 non-null  str
 2   Release date                122611 non-null  str
 3   Estimated owners            122611 non-null  str
 4   Peak CCU                    122611 non-null  int64
 5   Required age                122611 non-null  int64
 6   Price                       122611 non-null  float64
 7   DiscountDLC count           122611 non-null  int64
 8   About the game              122611 non-null  int64
 9   Supported languages         114162 non-null  str
 10  Full audio languages        122611 non-null  str
 11  Reviews                     122611 non-null  str
 12  Header image                12070 non-null   str
 13  Website                     122530 non-null  str
 14  Support url                 49676 non-null   str
 15  Support email               54142 non-null   str
 16  Windows                     100348 non-null  str
 17  Mac                         122611 non-null  bool
 18  Linux                       122611 non-null  bool
 19  Metacritic score            122611 non-null  bool
 20  Metacritic url              122611 non-null  int64
 21  User score                  4256 non-null    str
 22  Positive                    122611 non-null  int64
 23  Negative                    122611 non-null  int64
 24  Score rank                  122611 non-null  int64
 25  Achievements                40 non-null      float64
 26  Recommendations             122611 non-null  int64
 27  Notes                       122611 non-null  int64
 28  Average playtime forever    22458 non-null   str
 29  Average playtime two weeks  122611 non-null  int64
```

```
 30  Median playtime forever     122611 non-null  int64
 31  Median playtime two weeks   122611 non-null  int64
 32  Developers                  122611 non-null  int64
 33  Publishers                  114174 non-null  str
 34  Categories                  113702 non-null  str
 35  Genres                      113658 non-null  str
 36  Tags                        114198 non-null  str
 37  Screenshots                 83346 non-null   str
 38  Movies                      116593 non-null  str
 39  Column1                     0 non-null       float64
dtypes: bool(3), float64(3), int64(15), str(19)
memory usage: 35.0 MB
```

[88]:
```
dfgames.columns
```

[88]:
```
Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
       'Required age', 'Price', 'DiscountDLC count', 'About the game',
       'Supported languages', 'Full audio languages', 'Reviews',
       'Header image', 'Website', 'Support url', 'Support email', 'Windows',
       'Mac', 'Linux', 'Metacritic score', 'Metacritic url', 'User score',
       'Positive', 'Negative', 'Score rank', 'Achievements', 'Recommendations',
       'Notes', 'Average playtime forever', 'Average playtime two weeks',
       'Median playtime forever', 'Median playtime two weeks', 'Developers',
       'Publishers', 'Categories', 'Genres', 'Tags', 'Screenshots', 'Movies',
       'Column1'],
      dtype='str')
```

[89]:
```
dfgames2.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 13913 entries, 0 to 13912
Data columns (total 16 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   ID                   13913 non-null  int64
 1   Tittle               13913 non-null  str
 2   ReleaseDate          13884 non-null  str
 3   Rating               11610 non-null  str
 4   Genres               13913 non-null  str
 5   Description          13871 non-null  str
 6   Platforms            13912 non-null  str
 7   Metascore            13912 non-null  float64
 8   Metascore_count      13913 non-null  int64
 9   Metascore_sentiment  13912 non-null  str
 10  Userscore            13913 non-null  int64
 11  Userscore_count      13913 non-null  int64
 12  Userscore_sentiment  12440 non-null  str
 13  Platform_metascores  13912 non-null  str
```

```
 14  Developer            13904 non-null  str
 15  Publisher            13911 non-null  str
dtypes: float64(1), int64(4), str(11)
memory usage: 1.7 MB
```

[90]:
```
cols_validas = [
    "AppID", "Name", "Release date", "Estimated owners",
    "Peak CCU", "Required age", "Price",
    "Positive", "Negative", "Recommendations"
]

dfgames = dfgames[cols_validas]
```

[91]:
```
dfgames.shape
```

[91]:
```
(122611, 10)
```

[92]:
```
dfgames.columns
```

[92]:
```
Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
       'Required age', 'Price', 'Positive', 'Negative', 'Recommendations'],
      dtype='str')
```

[93]:
```
dfgames2.columns
```

[93]:
```
Index(['ID', 'Tittle', 'ReleaseDate', 'Rating', 'Genres', 'Description',
       'Platforms', 'Metascore', 'Metascore_count', 'Metascore_sentiment',
       'Userscore', 'Userscore_count', 'Userscore_sentiment',
       'Platform_metascores', 'Developer', 'Publisher'],
      dtype='str')
```

[94]:
```
dfgames2 = dfgames2.drop(columns=[
    'ID',
    'Description',
    'Platform_metascores',
    'Metascore_sentiment',
    'Userscore_sentiment',
    'Metascore_count',
    'Userscore_count'
])
```

[95]:
```
dfgames2.columns
```

[95]:
```
Index(['Tittle', 'ReleaseDate', 'Rating', 'Genres', 'Platforms', 'Metascore',
       'Userscore', 'Developer', 'Publisher'],
      dtype='str')
```

[96]:
```
dfgames.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 122611 entries, 0 to 122610
Data columns (total 10 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   AppID             122611 non-null  int64
 1   Name              122610 non-null  str
 2   Release date      122611 non-null  str
 3   Estimated owners  122611 non-null  str
 4   Peak CCU          122611 non-null  int64
 5   Required age      122611 non-null  int64
 6   Price             122611 non-null  float64
 7   Positive          122611 non-null  int64
 8   Negative          122611 non-null  int64
 9   Recommendations   122611 non-null  int64
dtypes: float64(1), int64(6), str(3)
memory usage: 9.4 MB
```

[97]:
```
dfgames.isnull().sum()
```

[97]:
```
AppID               0
Name                1
Release date        0
Estimated owners    0
Peak CCU            0
Required age        0
Price               0
Positive            0
Negative            0
Recommendations     0
dtype: int64
```

[98]:
```
dfgames.columns
```

[98]:
```
Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
       'Required age', 'Price', 'Positive', 'Negative', 'Recommendations'],
      dtype='str')
```

[99]:
```
dfgames.shape
```

[99]:
```
(122611, 10)
```

[100]:
```
dfgames["Release date"] = pd.to_datetime(dfgames["Release date"],␣
 ↪errors="coerce")
dfgames["Name"] = dfgames["Name"].str.strip().str.lower()
```

[101]:
```
dfgames2 = dfgames2.reset_index()
```

[102]:
```
dfgames.head()
```

[102]:
```
     AppID                                   Name Release date  \
0  2539430             black dragon mage playtest   2023-08-01
1   496350  supipara - chapter 1 spring has come!   2016-07-29
2  1034400      mystery solitaire the black raven   2019-05-06
3  3292190                   - vtuber paranoia   2024-10-31
4  3631080                           maze quest vr   2025-04-24

  Estimated owners  Peak CCU  Required age  Price  Positive  Negative  \
0            0 - 0         0             0   0.00         0         0
1        0 - 20000         0             0   5.24         0       252
2        0 - 20000         0             0   4.99         0        21
3        0 - 20000         1             0   8.99         0         0
4        0 - 20000         0             0   4.99         0         0

   Recommendations
0                0
1                0
2                0
3               19
4                0
```

[103]:
```
dfgames.isnull().sum()
```

[103]:
```
AppID               0
Name                1
Release date        0
Estimated owners    0
Peak CCU            0
Required age        0
Price               0
Positive            0
Negative            0
Recommendations     0
dtype: int64
```

[104]:
```
dfgames.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 122611 entries, 0 to 122610
Data columns (total 10 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   AppID             122611 non-null  int64
```

```
 1   Name               122610 non-null  str
 2   Release date       122611 non-null  datetime64[us]
 3   Estimated owners   122611 non-null  str
 4   Peak CCU           122611 non-null  int64
 5   Required age       122611 non-null  int64
 6   Price              122611 non-null  float64
 7   Positive           122611 non-null  int64
 8   Negative           122611 non-null  int64
 9   Recommendations    122611 non-null  int64
dtypes: datetime64[us](1), float64(1), int64(6), str(2)
memory usage: 9.4 MB
```

[105]:
```
dfgames["Price"].describe()
```

[105]:
```
count    122611.000000
mean          4.765091
std          12.531030
min           0.000000
25%           0.550000
50%           2.240000
75%           5.240000
max         999.980000
Name: Price, dtype: float64
```

[106]:
```
dfgames2.columns
```

[106]:
```
Index(['index', 'Tittle', 'ReleaseDate', 'Rating', 'Genres', 'Platforms',
       'Metascore', 'Userscore', 'Developer', 'Publisher'],
      dtype='str')
```

[107]:
```
dfgames2.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 13913 entries, 0 to 13912
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   index        13913 non-null  int64
 1   Tittle       13913 non-null  str
 2   ReleaseDate  13884 non-null  str
 3   Rating       11610 non-null  str
 4   Genres       13913 non-null  str
 5   Platforms    13912 non-null  str
 6   Metascore    13912 non-null  float64
 7   Userscore    13913 non-null  int64
 8   Developer    13904 non-null  str
 9   Publisher    13911 non-null  str
dtypes: float64(1), int64(2), str(7)
```

```
memory usage: 1.1 MB
```

[108]:
```
dfgames2.isnull().sum()
```

[108]:
```
index            0
Tittle           0
ReleaseDate     29
Rating        2303
Genres           0
Platforms        1
Metascore        1
Userscore        0
Developer        9
Publisher        2
dtype: int64
```

[109]:
```
dfgames2 = dfgames2.rename(columns={"index": "ID"})
```

[110]:
```
dfgames2.isnull().sum()
```

[110]:
```
ID               0
Tittle           0
ReleaseDate     29
Rating        2303
Genres           0
Platforms        1
Metascore        1
Userscore        0
Developer        9
Publisher        2
dtype: int64
```

[111]:
```
dfgames2["Rating"]=dfgames2["Rating"].fillna("Unknown")
dfgames2=dfgames2.
 ↪dropna(subset=["Platforms","Metascore","Developer","Publisher"])
```

[112]:
```
dfgames2.info()
```

```
<class 'pandas.DataFrame'>
Index: 13902 entries, 0 to 13911
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   ID           13902 non-null  int64
 1   Tittle       13902 non-null  str
 2   ReleaseDate  13873 non-null  str
 3   Rating       13902 non-null  str
 4   Genres       13902 non-null  str
```

```
 5   Platforms    13902 non-null  str
 6   Metascore    13902 non-null  float64
 7   Userscore    13902 non-null  int64
 8   Developer    13902 non-null  str
 9   Publisher    13902 non-null  str
dtypes: float64(1), int64(2), str(7)
memory usage: 1.2 MB
```

[113]:
```
dfgames2["ReleaseDate"] = pd.to_datetime(dfgames2["ReleaseDate"],␣
↪errors="coerce")
dfgames2 = dfgames2.rename(columns={"Tittle": "Title"})
dfgames2["Title"] = dfgames2["Title"].str.strip().str.lower()
dfgames2["Developer"] = dfgames2["Developer"].str.strip().str.lower()
dfgames2["Publisher"] = dfgames2["Publisher"].str.strip().str.lower()
```

[114]:
```
dfgames2 = dfgames2.drop_duplicates()
```

[115]:
```
dfgames2.info()
```

```
<class 'pandas.DataFrame'>
Index: 13902 entries, 0 to 13911
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   ID           13902 non-null  int64
 1   Title        13902 non-null  str
 2   ReleaseDate  13873 non-null  datetime64[us]
 3   Rating       13902 non-null  str
 4   Genres       13902 non-null  str
 5   Platforms    13902 non-null  str
 6   Metascore    13902 non-null  float64
 7   Userscore    13902 non-null  int64
 8   Developer    13902 non-null  str
 9   Publisher    13902 non-null  str
dtypes: datetime64[us](1), float64(1), int64(2), str(6)
memory usage: 1.2 MB
```

[116]:
```
dfgames2["Metascore"].describe()
```

[116]:
```
count    13902.000000
mean        70.591785
std         12.272827
min         11.000000
25%         64.000000
50%         72.000000
75%         79.000000
max         99.000000
```

```
Name: Metascore, dtype: float64
```

[117]:
```
dfgames2["Userscore"].describe()
```

[117]:
```
count    13902.000000
mean        61.887498
std         24.739244
min          0.000000
25%         58.000000
50%         70.000000
75%         78.000000
max        100.000000
Name: Userscore, dtype: float64
```

[118]:
```
dfgames2["Genres"].value_counts().head(10)
```

[118]:
```
Genres
Action Adventure      784
2D Platformer         703
Action RPG            688
FPS                   594
JRPG                  389
Real-Time Strategy    381
Open-World Action     368
Survival              352
Adventure             348
Point-and-Click       337
Name: count, dtype: int64
```

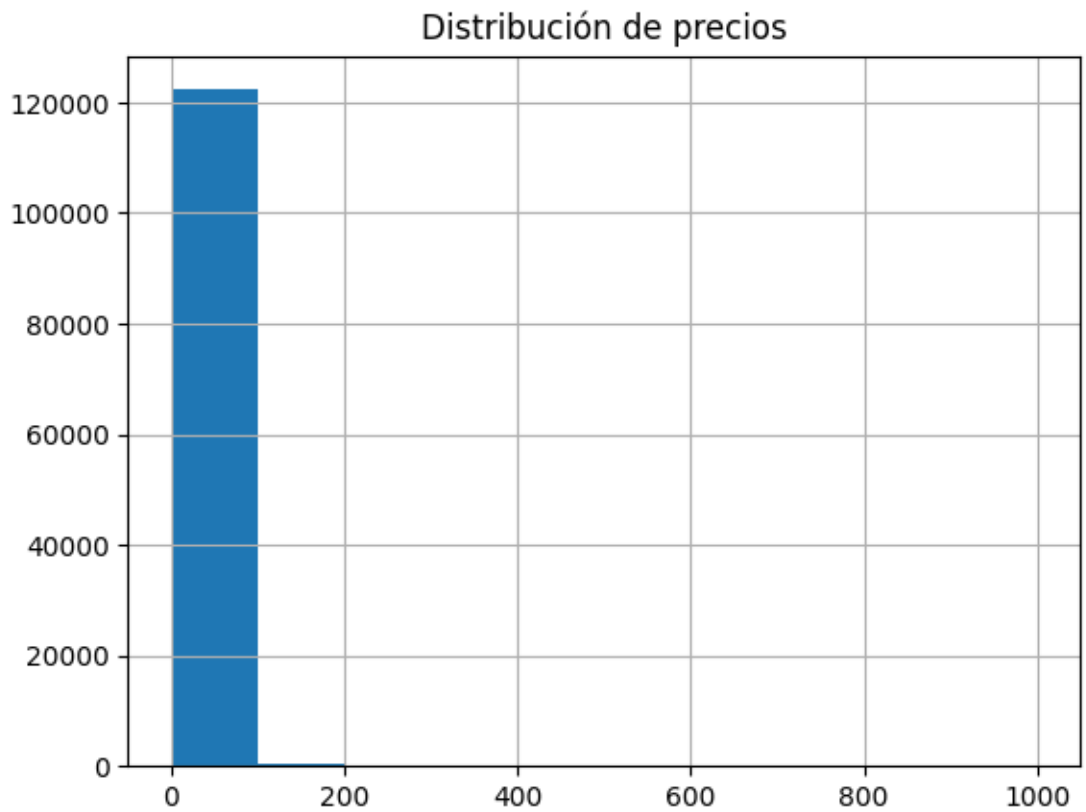
[119]:
```
dfgames2["Publisher"].value_counts().head(10)
```

[119]:
```
Publisher
nintendo                  479
ubisoft                   364
sega                      334
electronic arts           308
konami                    293
capcom                    255
microsoft game studios    254
square enix               246
activision                208
thq                       156
Name: count, dtype: int64
```

[120]:
```
dfgames2.groupby("Genres")["Metascore"].mean().sort_values(ascending=False)
```

[120]:
```
Genres
Soccer Management    78.852941
Metroidvania         77.921053
Soccer Sim           77.850746
Baseball Sim         77.409091
Football Sim         76.666667
                        …
Athletics            59.028571
Fishing              57.880000
Sports               57.115385
Hunting              55.000000
Bowling              54.500000
Name: Metascore, Length: 119, dtype: float64
```

[121]:
```python
import matplotlib.pyplot as plt

dfgames["Price"].hist()
plt.title("Distribución de precios")
plt.show()
```



[122]:
```python
dfgames["AppID"].info()
```

```
<class 'pandas.Series'>
RangeIndex: 122611 entries, 0 to 122610
Series name: AppID
Non-Null Count   Dtype
--------------   -----
122611 non-null  int64
dtypes: int64(1)
memory usage: 958.0 KB
```

[123]:
```
dfgames2["ID"].info()
```

```
<class 'pandas.Series'>
Index: 13902 entries, 0 to 13911
Series name: ID
Non-Null Count  Dtype
--------------  -----
13902 non-null  int64
dtypes: int64(1)
memory usage: 217.2 KB
```

[124]:
```
dfgames["rating"] = np.where(
    (dfgames["Positive"] + dfgames["Negative"]) > 0,
    dfgames["Positive"] / (dfgames["Positive"] + dfgames["Negative"]),
    np.nan
)
```

[125]:
```
dfgames["owners_min"] = dfgames["Estimated owners"].str.split("-").str[0].str.
 ↪replace(",", "").astype(int)
dfgames["owners_max"] = dfgames["Estimated owners"].str.split("-").str[1].str.
 ↪replace(",", "").astype(int)
```

[126]:
```
dfgames["year"] = dfgames["Release date"].dt.year
```

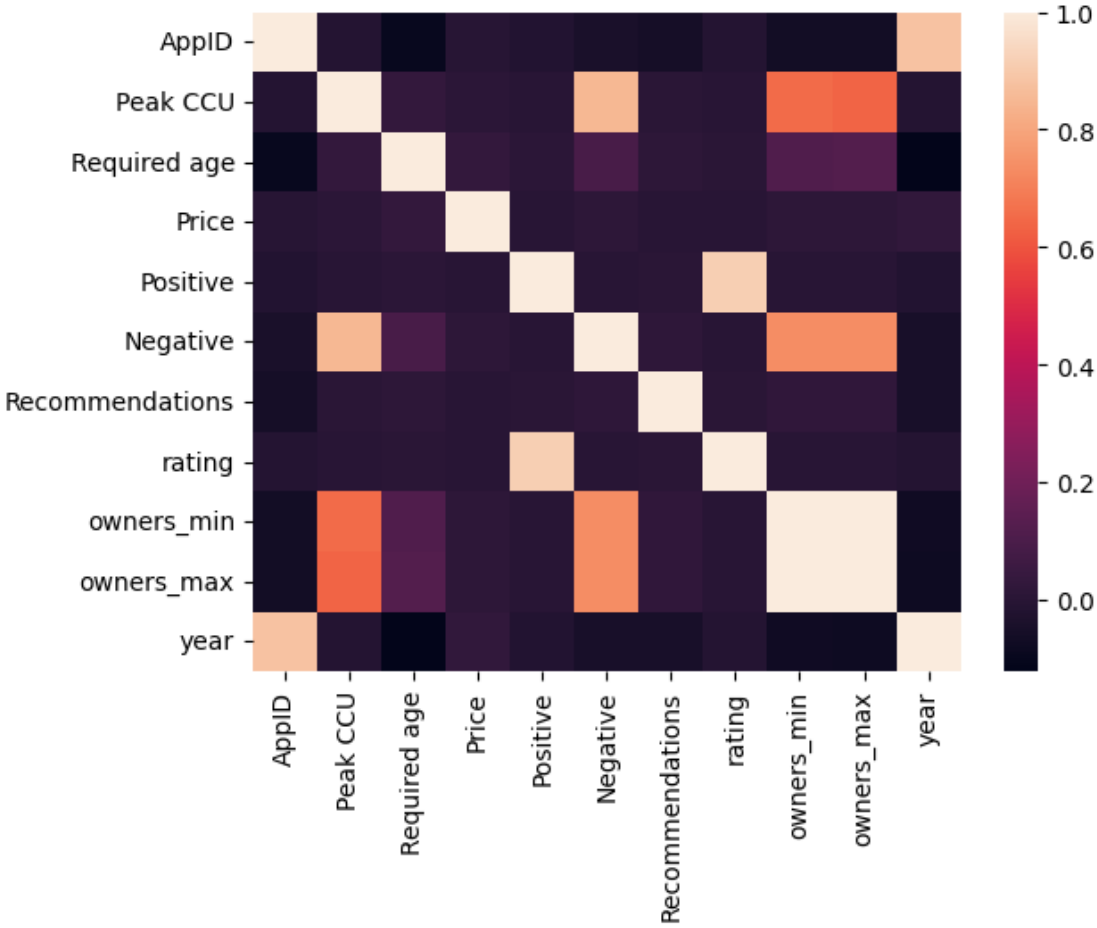
[127]:
```
dfgames.corr(numeric_only=True)
```

[127]:
```
                    AppID  Peak CCU  Required age     Price  Positive  \
AppID            1.000000 -0.014344     -0.096901 -0.007492 -0.018938
Peak CCU        -0.014344  1.000000      0.030635  0.004664 -0.000252
Required age    -0.096901  0.030635      1.000000  0.028655  0.008382
Price           -0.007492  0.004664      0.028655  1.000000 -0.000949
Positive        -0.018938 -0.000252      0.008382 -0.000949  1.000000
Negative        -0.043735  0.846709      0.090763  0.011233 -0.000592
Recommendations -0.059703  0.004201      0.010484 -0.002820  0.001651
rating          -0.016592 -0.000362      0.002994 -0.002127  0.916497
owners_min      -0.066667  0.651815      0.113270  0.012126 -0.000868
owners_max      -0.069650  0.635771      0.118740  0.013009 -0.000868
year             0.879929 -0.016259     -0.122647  0.023298 -0.019918
```

```
                  Negative  Recommendations    rating  owners_min  owners_max  \
AppID            -0.043735        -0.059703 -0.016592   -0.066667   -0.069650
Peak CCU          0.846709         0.004201 -0.000362    0.651815    0.635771
Required age      0.090763         0.010484  0.002994    0.113270    0.118740
Price             0.011233        -0.002820 -0.002127    0.012126    0.013009
Positive         -0.000592         0.001651  0.916497   -0.000868   -0.000868
Negative          1.000000         0.013942 -0.000920    0.734410    0.734566
Recommendations   0.013942         1.000000  0.000718    0.018618    0.019278
rating           -0.000920         0.000718  1.000000   -0.001458   -0.001476
owners_min        0.734410         0.018618 -0.001458    1.000000    0.996050
owners_max        0.734566         0.019278 -0.001476    0.996050    1.000000
year             -0.049554        -0.051535 -0.016494   -0.078048   -0.080852

                     year
AppID            0.879929
Peak CCU        -0.016259
Required age    -0.122647
Price            0.023298
Positive        -0.019918
Negative        -0.049554
Recommendations -0.051535
rating          -0.016494
owners_min      -0.078048
owners_max      -0.080852
year             1.000000
```

[128]:
```python
import seaborn as sns
sns.heatmap(dfgames.corr(numeric_only=True))
```

[128]: <Axes: >



[129]:
```
dfgames.groupby(pd.cut(dfgames["Price"], bins=5))["Recommendations"].mean()
```

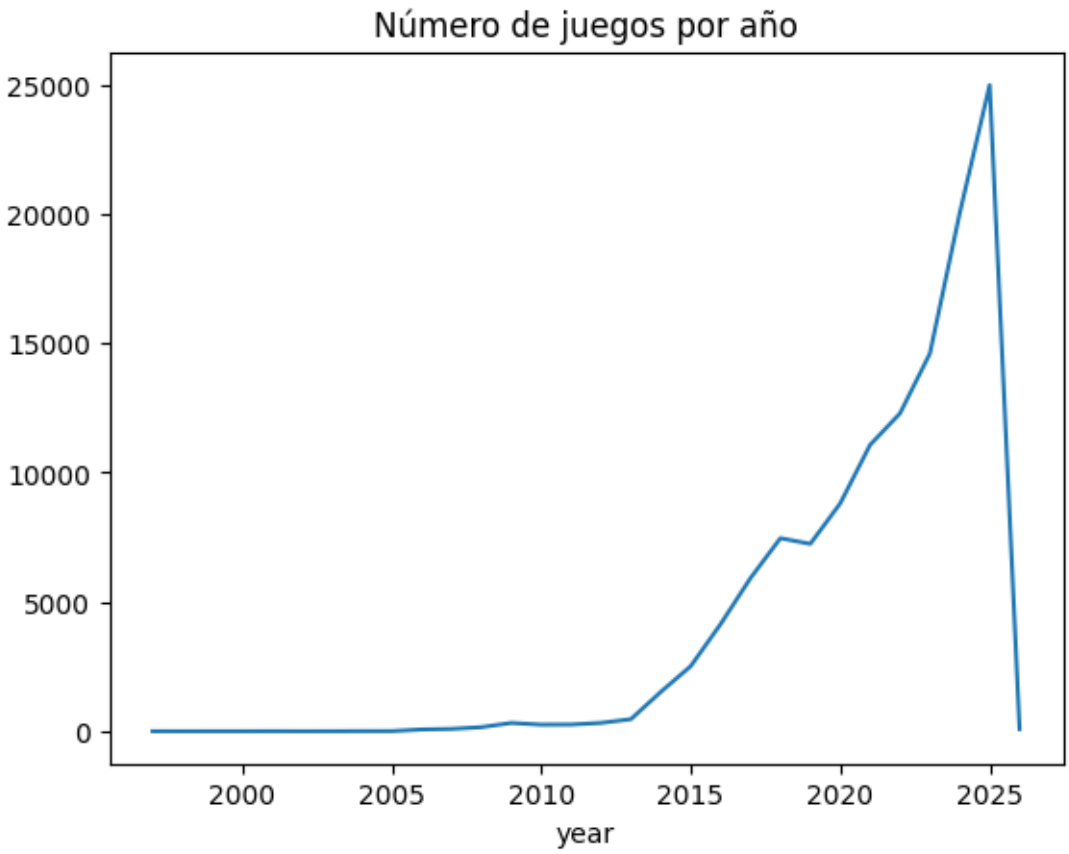
[129]:
```
Price
(-1.0, 199.996]       18.088048
(199.996, 399.992]     0.000000
(399.992, 599.988]     0.000000
(799.984, 999.98]      0.000000
Name: Recommendations, dtype: float64
```

[130]:
```
dfgames.sort_values("Peak CCU", ascending=False).head(10)
```

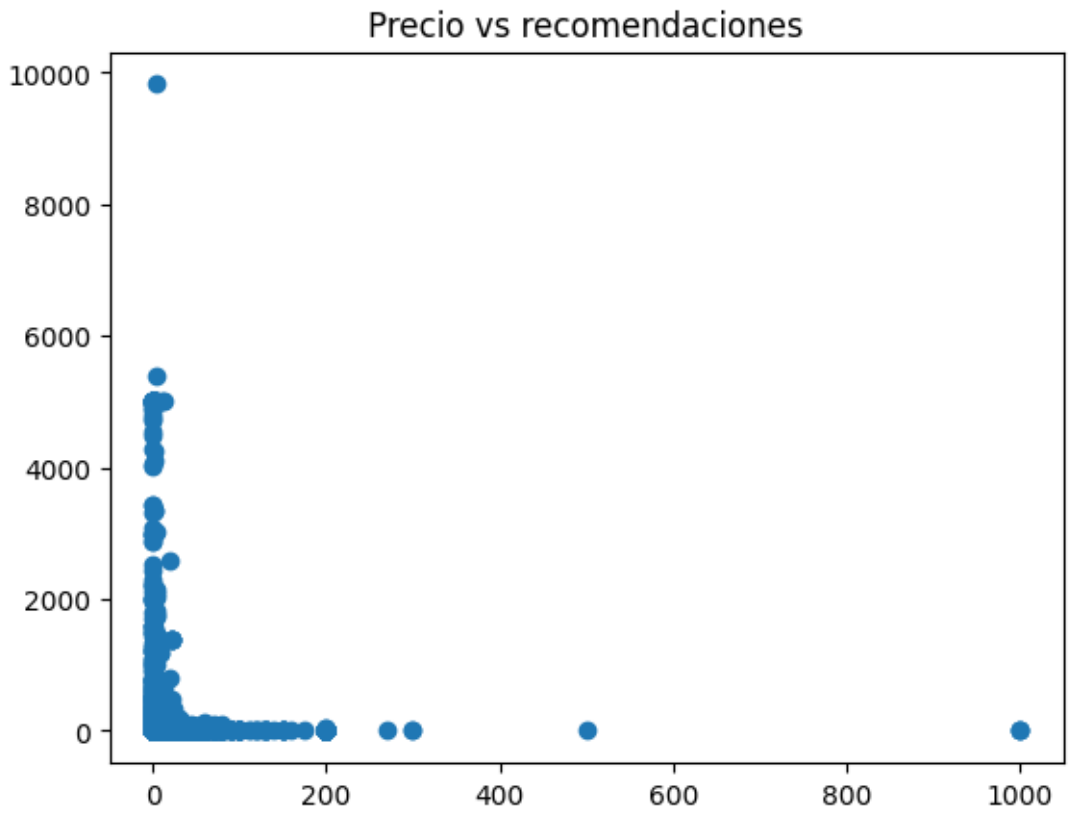
[130]:
```
          AppID                   Name Release date       Estimated owners  \
45509       730       counter-strike 2   2012-08-21  100000000 - 200000000
4193        570                 dota 2   2013-07-09  100000000 - 200000000
8878     578080    pubg: battlegrounds   2017-12-21  100000000 - 200000000
104347  3419430              bongo cat   2025-03-05              0 - 20000
17581   2622380  elden ring nightreign   2025-05-29      2000000 - 5000000
5377     252490                   rust   2018-02-08    20000000 - 50000000
```

```
54502   1172470          apex legends    2020-11-04  100000000 - 200000000
109015  2767030          marvel rivals   2024-12-05              0 - 20000
86168    431960       wallpaper engine   2018-11-16    20000000 - 50000000
15313   1671210              deltarune   2025-06-04      1000000 - 2000000

        Peak CCU  Required age  Price  Positive  Negative  Recommendations  \
45509    1013936             0   0.00         0   7642084                1
4193      623941             0   0.00         0   2037143                0
8878      314682            13   0.00         0   1520457               37
104347    164168             0   0.00         0         0                7
17581     163599            17  29.99         0     48901               37
5377      143870             0  19.99         0   1071135               92
54502     124262             0   0.00         0    668053               12
109015    106611             0   0.00         0    228817               49
86168      91184             0   3.99         0    876898               17
15313      83936             0  19.99         0      2804                0

        rating  owners_min  owners_max  year
45509      0.0   100000000   200000000  2012
4193       0.0   100000000   200000000  2013
8878       0.0   100000000   200000000  2017
104347     NaN           0       20000  2025
17581      0.0     2000000     5000000  2025
5377       0.0    20000000    50000000  2018
54502      0.0   100000000   200000000  2020
109015     0.0           0       20000  2024
86168      0.0    20000000    50000000  2018
15313      0.0     1000000     2000000  2025
```

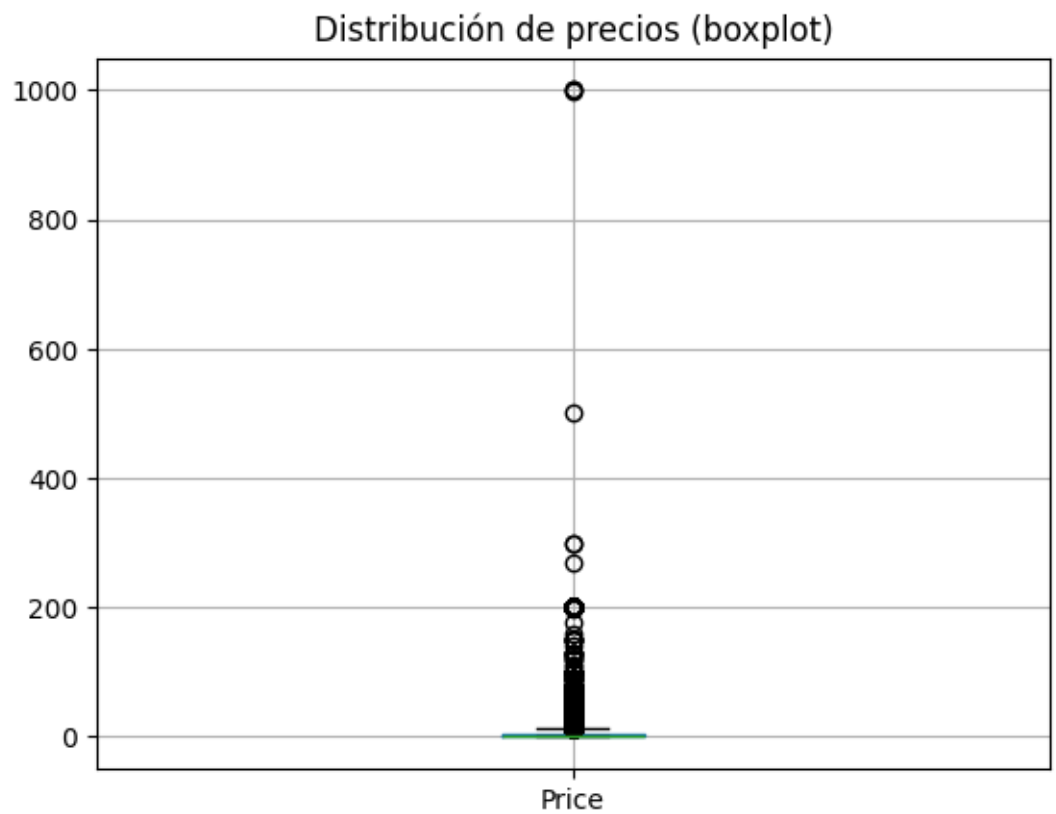
[131]:
```python
dfgames["year"].value_counts().sort_index().plot()
plt.title("Número de juegos por año")
plt.show()
```



[132]:
```
plt.scatter(dfgames["Price"], dfgames["Recommendations"])
plt.title("Precio vs recomendaciones")
plt.show()
```



[133]:
```
dfgames.boxplot(column="Price")
plt.title("Distribución de precios (boxplot)")
plt.show()
```



[134]:
```
dfgames["Name_clean"] = dfgames["Name"].str.lower().str.strip()
dfgames2["Title_clean"] = dfgames2["Title"].str.lower().str.strip()
```

[135]:
```
df_final = dfgames.merge(
    dfgames2,
    left_on="Name_clean",
    right_on="Title_clean",
    how="left"
)

match_rate = df_final["Metascore"].notnull().mean()
print("Match rate:", match_rate)
```

```
Match rate: 0.03867515965125478
```

[136]:
```
df_final.groupby("Genres")["Metascore"].mean().sort_values(ascending=False).
 ↪head(10)
```

[136]:
```
Genres
Exercise              89.000000
Baseball              83.000000
```

```
Basketball Sim       82.666667
Card Battle          79.790698
Text Adventure       77.800000
Logic Puzzle         77.750000
Arcade Racing        77.714286
Metroidvania         77.618182
Compilation          77.533333
Soccer Management    77.500000
Name: Metascore, dtype: float64
```

[137]:
```
df_final = df_final.drop(columns=["Name_clean", "Title_clean"])
```

[138]:
```
df_final.groupby("year")["Recommendations"].mean().plot()
plt.title("Recomendaciones medias por año")
plt.show()
```

[139]:
```
df_final["is_free"] = df_final["Price"] == 0

df_final.groupby("is_free")["Recommendations"].mean()
```

[139]:
```
is_free
False    20.918251
True      7.671640
Name: Recommendations, dtype: float64
```

[140]:
```python
df_final.sort_values("Recommendations", ascending=False)[["Name",␣
↪"Recommendations"]].head(10)
```

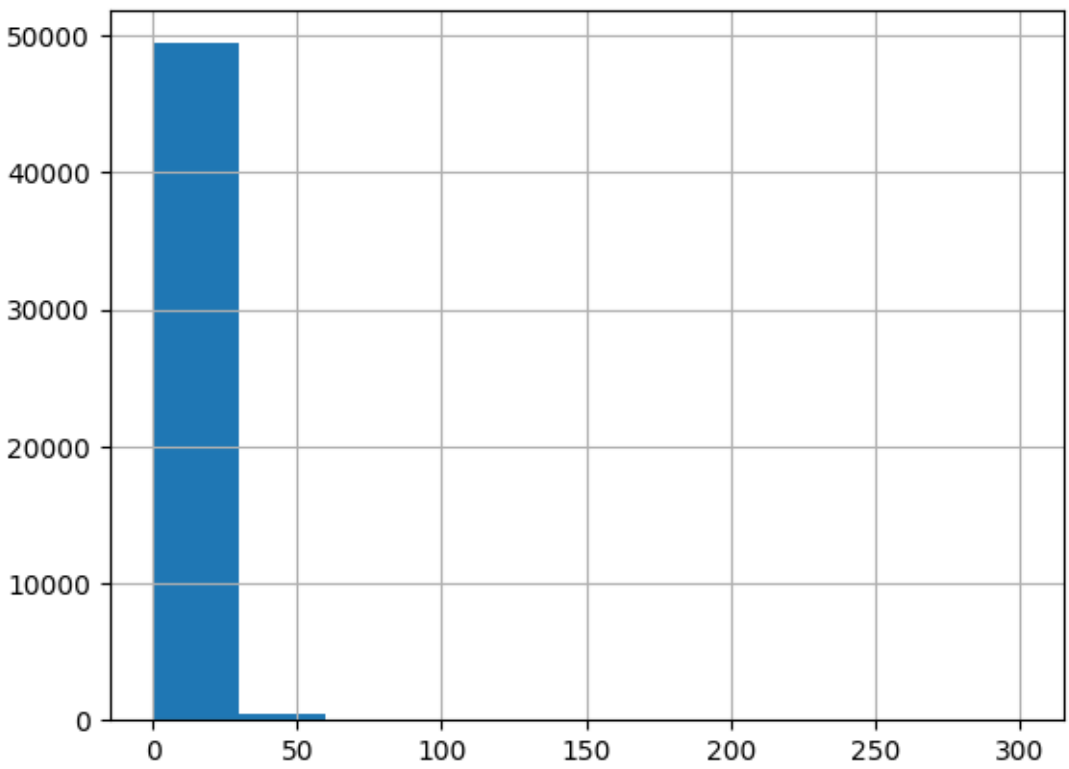
[140]:

	Name	Recommendations
76082	logistical	9821
8313	drive-by hero	5394
114275	trivia vault: super heroes trivia 2	5000
90761	trivia vault: mixed trivia	5000
115240	ninja stealth	5000
108803	trivia vault: auto racing trivia	5000
29400	professor watts word search: into the ocean	5000
1241	trivia vault: classic rock trivia 2	5000
108587	trivia vault: classic rock trivia	5000
42012	trivia vault football trivia	5000

[141]:
```python
df_final = df_final[(df_final["Positive"] + df_final["Negative"]) > 10]
```

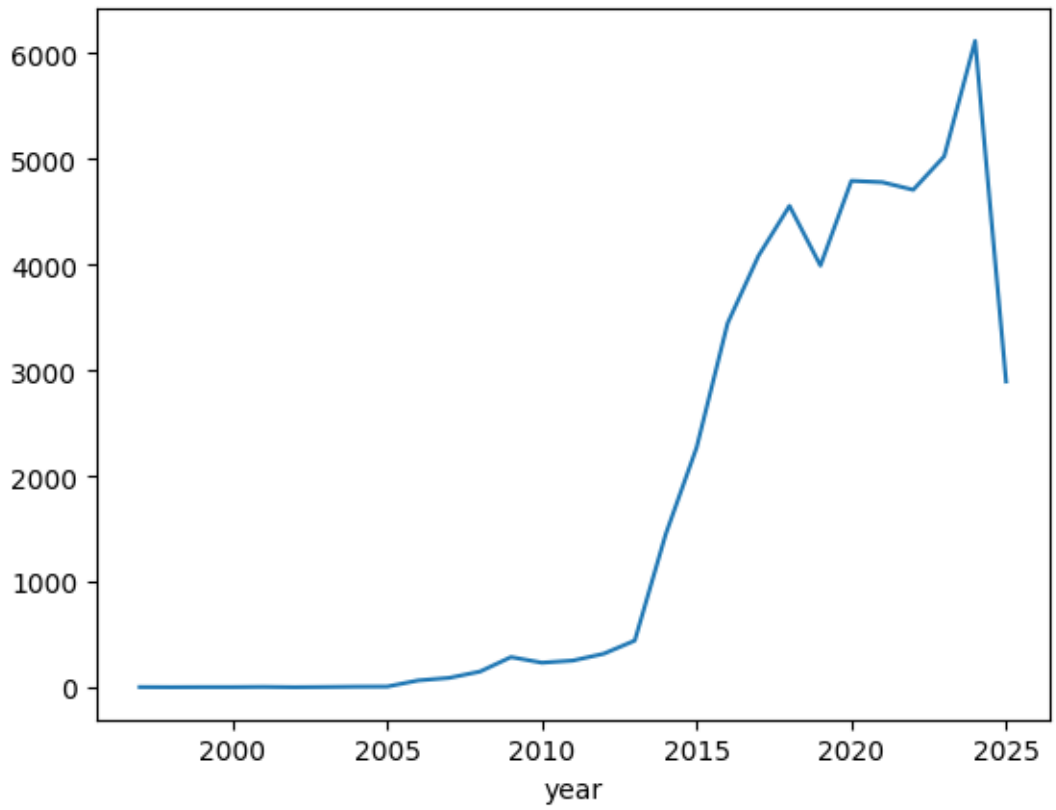
[142]:
```python
df_final.describe()
df_final["Price"].hist()
```

[142]: <Axes: >



[143]:
```
df_final["year"].value_counts().sort_index().plot()
```

[143]: <Axes: xlabel='year'>



[144]:
```
print("Total registros:", len(df_final))
print("Con Metascore:", df_final["Metascore"].notnull().sum())
print("Sin Metascore:", df_final["Metascore"].isnull().sum())
```

```
Total registros: 49987
Con Metascore: 4450
Sin Metascore: 45537
```

[145]:
```
df_final = df_final[df_final["Price"] < 100]
```

[146]:
```
df_final["Price"].describe()
```

[146]:
```
count    49937.000000
mean         4.924320
std          6.320743
min          0.000000
25%          0.990000
50%          2.990000
75%          6.490000
max         99.990000
Name: Price, dtype: float64
```

[147]:
```
df_final["total_reviews"] = df_final["Positive"] + df_final["Negative"]

df_final["rating"] = np.where(
    df_final["total_reviews"] > 0,
    df_final["Positive"] / df_final["total_reviews"],
    np.nan
)
```

[148]:
```
df_final = df_final[df_final["total_reviews"] > 20]
```

[149]:
```
df_final.sort_values("Peak CCU", ascending=False).head(10)
```

[149]:
```
          AppID                       Name Release date  \
45509       730          counter-strike 2   2012-08-21
4193        570                     dota 2   2013-07-09
8878     578080        pubg: battlegrounds   2017-12-21
17581   2622380      elden ring nightreign   2025-05-29
5377     252490                       rust   2018-02-08
54502   1172470              apex legends   2020-11-04
109015  2767030              marvel rivals   2024-12-05
86168    431960          wallpaper engine   2018-11-16
15313   1671210                 deltarune   2025-06-04
59122    271590  grand theft auto v legacy   2015-04-13

             Estimated owners  Peak CCU  Required age  Price  Positive  \
45509   100000000 - 200000000   1013936             0   0.00         0
4193    100000000 - 200000000    623941             0   0.00         0
8878    100000000 - 200000000    314682            13   0.00         0
17581       2000000 - 5000000    163599            17  29.99         0
5377      20000000 - 50000000    143870             0  19.99         0
54502   100000000 - 200000000    124262             0   0.00         0
109015              0 - 20000    106611             0   0.00         0
86168     20000000 - 50000000     91184             0   3.99         0
15313       1000000 - 2000000     83936             0  19.99         0
59122    50000000 - 100000000     67851            17   0.00         0

        Negative  Recommendations  …  ReleaseDate   Rating  \
45509    7642084                1  …   2023-09-27  Unknown
4193     2037143                0  …   2013-07-09  Unknown
8878     1520457               37  …          NaT      NaN
17581      48901               37  …   2025-05-30        M
5377     1071135               92  …   2018-02-08  Unknown
54502     668053               12  …          NaT      NaN
109015    228817               49  …   2024-12-06  Unknown
86168     876898               17  …          NaT      NaN
15313       2804                0  …   2025-06-05        T
59122    1739980               77  …          NaT      NaN
```

```
                                Genres                        Platforms  \
45509                     Tactical FPS                               PC
4193                              MOBA                               PC
8878                               NaN                              NaN
17581                       Action RPG  PlayStation 5,Xbox Series X,PC
5377                          Survival       PC,Xbox One,PlayStation 4
54502                              NaN                              NaN
109015  Tactical Third Person Shooter  PC,PlayStation 5,Xbox Series X
86168                              NaN                              NaN
15313                             JRPG                Nintendo Switch 2
59122                              NaN                              NaN

        Metascore Userscore          Developer  \
45509        82.0      53.0     valve software
4193         90.0      65.0     valve software
8878          NaN       NaN                NaN
17581        77.0      67.0      from software
5377         69.0      65.0  facepunch studios
54502         NaN       NaN                NaN
109015       74.0      69.0      netease games
86168         NaN       NaN                NaN
15313        86.0      90.0                8-4
59122         NaN       NaN                NaN

                                Publisher is_free total_reviews
45509                      valve software    True       7642084
4193                valve software,nexon    True       2037143
8878                                 NaN    True       1520457
17581   bandai namco games,from software   False         48901
5377                   facepunch studios   False       1071135
54502                                NaN    True        668053
109015                     netease games    True        228817
86168                                NaN   False        876898
15313                                8-4   False          2804
59122                                NaN    True       1739980

[10 rows x 26 columns]
```

[150]:
```
df_final["rating"].describe()
```

[150]:
```
count    38912.000000
mean         0.000625
std          0.021085
min          0.000000
25%          0.000000
50%          0.000000
```

```
75%          0.000000
max          0.909091
Name: rating, dtype: float64
```

[151]:
```python
df_final.sort_values("rating", ascending=False).head(10)
```

[151]:
```
          AppID                                  Name Release date  \
48515    471720                             k station   2016-06-15
65134    891870                        king of phoenix   2018-09-27
93790    929290           strip breaker : hentai girls   2018-09-30
27656    961640               the tower of five hearts   2018-11-14
118363   929310  kamasutra connect : sexy hentai girls   2018-11-13
38005    946800                           hentai 2+2=4   2018-10-05
51398   1013180                        funbag fantasy   2019-01-31
71761    966460                     undress tournament   2019-01-24
22730    677730                             karmasutra   2017-09-29
83974   1014730                                  cyndy   2019-05-16

       Estimated owners  Peak CCU  Required age  Price  Positive  Negative  \
48515         0 - 20000         0             0   0.00       100        10
65134         0 - 20000         0             0   2.49        84        11
93790         0 - 20000         0             0   0.49        53         8
27656         0 - 20000         0             0   7.49       100        16
118363        0 - 20000         0             0   0.49        68        11
38005         0 - 20000         0             0   0.49        61        10
51398         0 - 20000         1             0  29.24       100        21
71761         0 - 20000         0             0   0.00        57        12
22730         0 - 20000         0             0   8.99        60        13
83974         0 - 20000         0             0   7.99        68        16

        Recommendations  …  ReleaseDate  Rating  Genres  Platforms  \
48515                 0  …          NaT     NaN     NaN        NaN
65134                10  …          NaT     NaN     NaN        NaN
93790                 0  …          NaT     NaN     NaN        NaN
27656                 0  …          NaT     NaN     NaN        NaN
118363               18  …          NaT     NaN     NaN        NaN
38005                16  …          NaT     NaN     NaN        NaN
51398                 0  …          NaT     NaN     NaN        NaN
71761                10  …          NaT     NaN     NaN        NaN
22730                36  …          NaT     NaN     NaN        NaN
83974                21  …          NaT     NaN     NaN        NaN

        Metascore Userscore Developer Publisher is_free total_reviews
48515         NaN       NaN       NaN       NaN    True           110
65134         NaN       NaN       NaN       NaN   False            95
93790         NaN       NaN       NaN       NaN   False            61
27656         NaN       NaN       NaN       NaN   False           116
```

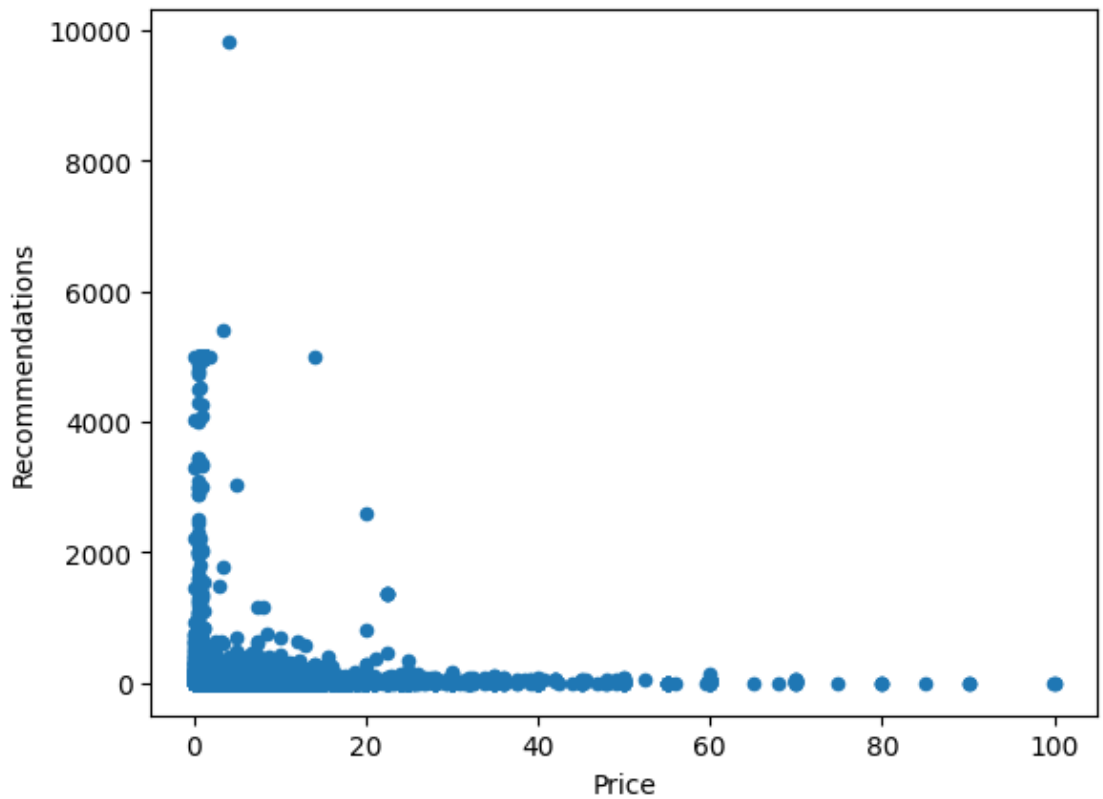
```
118363        NaN       NaN       NaN       NaN   False            79
38005         NaN       NaN       NaN       NaN   False            71
51398         NaN       NaN       NaN       NaN   False           121
71761         NaN       NaN       NaN       NaN    True            69
22730         NaN       NaN       NaN       NaN   False            73
83974         NaN       NaN       NaN       NaN   False            84

[10 rows x 26 columns]
```

[152]:
```
df_final = df_final[(df_final["Positive"] + df_final["Negative"]) > 10]
```

[ ]:
```

```

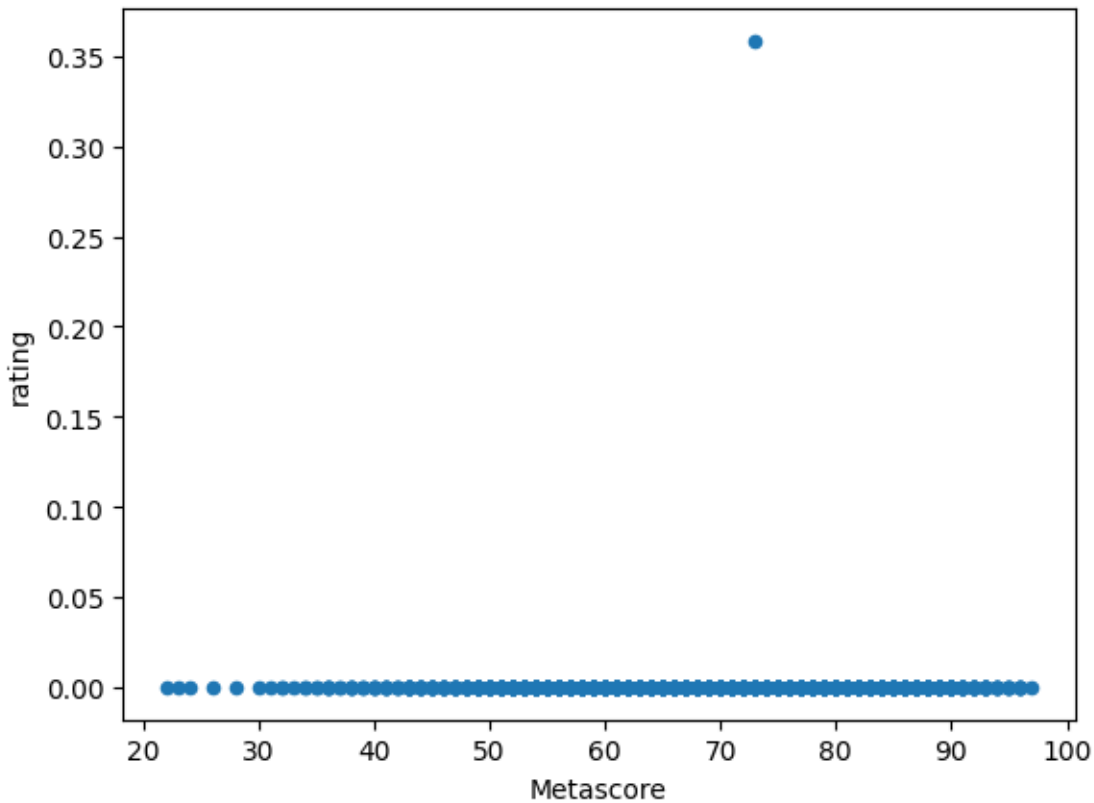
[153]:
```
df_final.plot.scatter(x="Price", y="Recommendations")
```

[153]: <Axes: xlabel='Price', ylabel='Recommendations'>



[154]:
```
df_final.plot.scatter(x="Metascore", y="rating")
```

[154]: <Axes: xlabel='Metascore', ylabel='rating'>

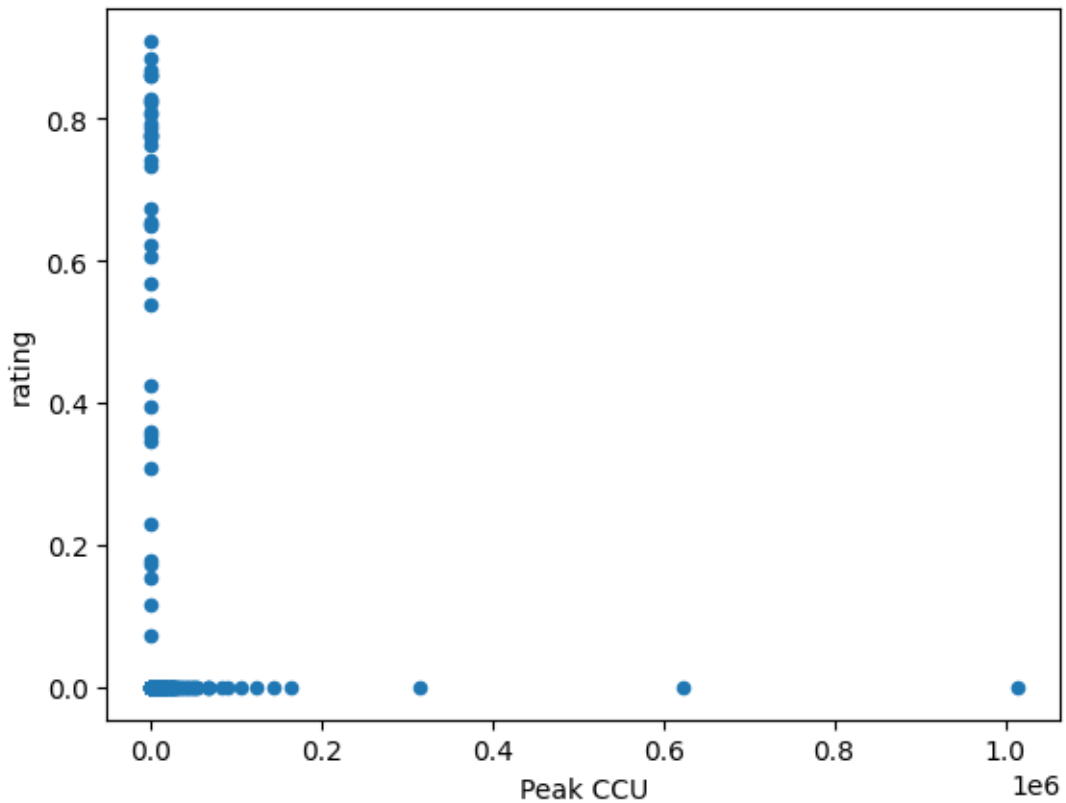


[155]:
```
df_final.groupby(df_final["Price"] == 0)["Recommendations"].mean()
```

[155]:
```
Price
False    39.238078
True     19.287775
Name: Recommendations, dtype: float64
```

[156]:
```
df_final.plot.scatter(x="Peak CCU", y="rating")
```

[156]:
```
<Axes: xlabel='Peak CCU', ylabel='rating'>
```



[157]:
```
df_final.groupby("year")["rating"].mean()
```

[157]:
```
year
1997    0.000000
1998    0.000000
1999    0.000000
2000    0.000000
2001    0.000000
2002    0.000000
2003    0.000000
2004    0.000000
2005    0.000000
2006    0.000000
2007    0.000000
2008    0.000000
2009    0.000000
2010    0.000000
2011    0.000000
2012    0.000000
2013    0.000000
2014    0.000000
```

```
2015    0.000148
2016    0.000787
2017    0.001187
2018    0.004029
2019    0.001146
2020    0.000048
2021    0.000000
2022    0.000000
2023    0.000000
2024    0.000000
2025    0.000000
Name: rating, dtype: float64
```

[158]:
```
df_final.to_csv("df_final.csv", index=False)
```

[159]:
```
df_final = df_final[(df_final["Positive"] + df_final["Negative"]) > 10]
```

[160]:
```
df_final.groupby("Genres")["Metascore"].mean().sort_values(ascending=False).
 ↪head(10)
```

[160]:
```
Genres
Exercise             89.000000
Baseball             83.000000
Basketball Sim       83.000000
Card Battle          79.914286
Sports               78.500000
Soccer Management    78.363636
Text Adventure       77.800000
Compilation          77.716981
Metroidvania         77.643564
MOBA                 77.444444
Name: Metascore, dtype: float64
```

[161]:
```
df_final["rating"] = np.where(
    (df_final["Positive"] + df_final["Negative"]) > 0,
    df_final["Positive"] / (df_final["Positive"] + df_final["Negative"]),
    np.nan
)

df_final = df_final[df_final["rating"] > 0]
```

[163]:
```
free = df_final[df_final["is_free"] == True]["Recommendations"]
paid = df_final[df_final["is_free"] == False]["Recommendations"]

print("Media free:", free.mean())
print("Media paid:", paid.mean())
print("Diferencia absoluta:", paid.mean() - free.mean())
```

```python
print("Ratio (paid/free):", paid.mean() / free.mean())
```

```
Media free: 6.111111111111111
Media paid: 40.61290322580645
Diferencia absoluta: 34.501792114695334
Ratio (paid/free): 6.64574780058651
```